

RESEARCH ARTICLE

Quantifying LLM Non-Determinism Across a Full Evidence-Synthesis Pipeline: A Multi-Repetition, Multi-Stage Reproducibility Study

Lucas Rover¹ and Yara de Souza Tadano¹

¹Programa de Pós-Graduação em Sustentabilidade Ambiental Urbana, Universidade Tecnológica Federal do Paraná (UTFPR), Curitiba, PR, Brazil. E-mail: lucasrover@alunos.utfpr.edu.br.

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Abstract

Background: Large language models (LLMs) are increasingly used in evidence synthesis, yet a critical assumption—that identical inputs produce identical outputs—remains largely untested across a complete synthesis pipeline. **Methods:** We quantified the reproducibility of six LLMs (three local: LLaMA 3 8B, Mistral 7B, Gemma 2 9B; three cloud APIs: Claude Sonnet 4.5, Gemini 2.5 Pro, GPT-4.1) across a two-stage pipeline—abstract screening and data extraction—applied to a 500-abstract PubMed corpus on fine particulate matter (PM_{2.5}) and respiratory health. Each model was run 10 times under identical configurations (temperature=0, fixed seeds where supported), totalling 35,996 successful calls of 36,000 attempted (99.99% success rate). To isolate deployment-vs.-model-size confounding, we additionally served `meta-llama/llama-3-8b-instruct` via a cloud endpoint (DeepInfra) under identical settings. We pre-committed analysis plans for the post-revision experiments to the public Git repository before result computation. The primary outcome was the Exact Match Rate (EMR) with 95% bootstrap confidence intervals. **Results:** Cloud commercial APIs achieved extraction EMR as low as 0.050 (Claude) despite temperature=0 and fixed seeds, while local open-weight models achieved EMR=1.000—an approximately 19× gap. Critically, the same LLaMA 3 8B model weights served via the cloud (DeepInfra) produced 39% systematic disagreement with the local Ollama deployment, refuting the model-size confound: deployment infrastructure alone—not parameter count—drives the gap. Across all 100 INCLUDE articles pooled by random-effects meta-analysis, the pooled estimate was robust to non-determinism; in random k=10 subsamples, however, 0.5% of subsamples produced run-dependent reversals of the meta-analytic conclusion. A pairwise-disagreement analysis ($\binom{10}{2}=45$ pairs per item) revealed that 95% of Claude’s extraction articles produced at least one differing run pair. BERTScore F1 on metadata saturated near 0.997, masking true narrative variation (Jaccard similarity on rationales ranged 0.38–0.73 for cloud APIs, vs. 0.98 on metadata). **Conclusions:** Cloud API non-determinism poses a quantifiable threat to evidence synthesis reproducibility, particularly for quantitative extraction tasks and small-k literatures. Single-run results should not be treated as definitive; multiple repetitions, provenance tracking, and—where feasible—deterministic local deployment should become standard practice.

1. Introduction

1.1. The Promise of AI in Evidence Synthesis

Evidence synthesis—the systematic process of identifying, appraising, and combining research findings—forms the backbone of evidence-based medicine and public health policy. Systematic reviews and meta-analyses sit atop the hierarchy of evidence, informing clinical guidelines,

regulatory decisions, and resource allocation worldwide.^{1,2} Yet the process is notoriously labor-intensive: a typical systematic review requires 6–18 months and hundreds of person-hours, with title and abstract screening alone consuming an estimated 25% of total effort.³

The emergence of large language models (LLMs) has created unprecedented opportunities to accelerate this process. Recent studies demonstrate that models such as GPT-4,⁴ Claude,⁵ and Gemini⁶ can screen abstracts with sensitivity approaching that of human reviewers,^{7,8} extract structured data from published studies with over 90% accuracy,^{9,10} and even assist in evidence quality assessment.^{11,12} A growing body of work has established LLM-assisted evidence synthesis as a legitimate and rapidly expanding methodological frontier.^{13–16} The 2025 Cambridge *Research Synthesis Methods* editorial¹⁷ explicitly invites methodological work characterizing the reproducibility of generative-AI-assisted evidence synthesis, including specific reporting requirements for prompt design, random-seed control, and external validation. The Cochrane 2024 guidance for AI-assisted screening and data extraction¹⁸ further requires that AI-assisted SR tools report inter-run agreement statistics—which the paired EMR and pairwise-disagreement reporting in this paper supplies.

1.2. The Hidden Problem: Non-Determinism

Beneath this promise lies a rarely examined assumption: *if you ask the same question twice, you should get the same answer*. In the context of LLM-assisted evidence synthesis, this means that running the same model with the same prompt, temperature, and random seed on the same abstract should produce identical screening decisions and extraction outputs every time.

This assumption does not hold in practice.

Recent work has demonstrated that commercial LLM APIs exhibit non-deterministic behavior even when configured for maximum reproducibility. Atil et al.¹⁹ showed accuracy variations of up to 15% across repeated identical runs, with best-to-worst performance gaps reaching 70%. Klishevich et al.²⁰ confirmed that temperature=0 does not guarantee deterministic outputs across any of the major models tested (GPT-4o, Claude 3.5, LLaMA 3.2). At a deeper level, Fu et al.²¹ demonstrated that even when surface-level text appears identical, the underlying token probability distributions exhibit substantial instability, suggesting that apparent determinism may mask hidden variation. A philosophical analysis by Karelin et al.²² traced this non-determinism to three fundamental sources: numerical precision in floating-point operations, architectural choices in distributed inference, and sampling procedures inherent to the generation process.

1.3. Why This Matters for Evidence Synthesis

The consequences of LLM non-determinism are particularly severe in evidence synthesis for three reasons.

First, decisions propagate. A screening decision that flips from “include” to “exclude” in a repeated run does not merely change one data point—it removes an entire study from downstream analysis. In meta-analysis, where pooled effect estimates are sensitive to study composition, a single inclusion change can shift a confidence interval across the null, transforming a statistically significant finding into a non-significant one.²³

Second, extraction compounds variation. Data extraction involves free-text parsing of complex scientific results—relative risks, confidence intervals, exposure definitions, population characteristics. Even small variations in how an LLM parses “RR=1.05 (95% CI: 1.01–1.09)” can propagate through meta-analytic pooling. Gartlehner et al.⁹ reported 95–97% test-retest reliability in LLM extraction, but this still implies 3–5% of extracted values may change between runs—a rate that, across hundreds of data points, can materially alter pooled estimates.

Third, reproducibility is the foundation of scientific credibility. The broader reproducibility crisis in AI research^{24–26} has already eroded trust in computational findings. When LLM-assisted reviews cannot be independently reproduced because the API returns different outputs, the scientific value of those reviews is undermined.^{27–29}

1.4. The Gap We Address

Despite the rapid adoption of LLMs in evidence synthesis, **no study has systematically measured how API-level non-determinism propagates through a complete synthesis pipeline**—from abstract screening through data extraction—in a real health domain. This is a critical blind spot: the field is deploying LLMs at scale on the assumption that outputs are stable, without evidence to support that assumption.

Consider the state of current evidence. Jensen et al.¹⁰ report 94.1% inter-session reproducibility for ChatGPT-4o extraction—an encouraging number. But this was measured with only 2 repetitions and evaluated per-field, not per-output. As we demonstrate below, applying a stricter whole-output metric across 10 repetitions yields extraction EMR as low as 5.0%—approximately 19× lower than what per-field metrics suggest. Similarly, Gartlehner et al.⁹ report 95–97% test-retest reliability, but with only 2–3 repetitions and no examination of how variation compounds across pipeline stages. Studies measuring non-determinism^{19,20} do so on general NLP benchmarks, where the structured, multi-field outputs characteristic of evidence synthesis are absent. And the most influential screening evaluations^{7,8,15} report results from a single run, implicitly assuming that a repeat would yield the same result.

We address this gap with the following contributions:

1. Quantifying reproducibility across 10 repeated runs of 6 LLMs (3 local, 3 API) through a two-stage evidence synthesis pipeline;
2. Measuring variation at multiple granularities: whole-output, per-field, per-abstract, and run-pair;
3. **Isolating the deployment effect from the model-size confound** via an additional 10-run experiment in which the same `meta-llama/llama-3-8b-instruct` model is served via a cloud endpoint (DeepInfra) under settings identical to the local Ollama deployment;
4. Testing whether prompt design (variable-length array vs. fixed-slot single-estimate) is the primary driver of cloud-API extraction non-determinism;
5. Cross-validating extraction outputs against two independent reference standards: a within-study majority-vote “silver-internal” and an external silver from DeepSeek-R1 (a reasoning model from a different family);
6. Demonstrating that common “deterministic” API settings (temperature=0, fixed seeds) are insufficient to ensure reproducibility, and that the seed parameter empirically delivers a 6× smaller effect than the deployment paradigm.

2. Related Work

2.1. LLMs in Systematic Reviews

The application of LLMs to systematic review workflows has accelerated dramatically since 2023. Oami et al.⁷ evaluated GPT-4 Turbo for citation screening, reporting sensitivity of 0.75 and specificity of 0.99, with prompt tuning improving sensitivity to 0.91. Khraisha et al.⁸ extended this to multilingual screening and extraction, finding strong specificity (>0.80) but variable sensitivity across languages and review types.

For data extraction, Gartlehner et al.⁹ showed that Claude 2 achieved 96.3% accuracy compared to human reviewers, with test-retest reliability of 95–97%. Jensen et al.¹⁰ found

that ChatGPT-4o achieves 92.4% extraction accuracy and 94.1% inter-session reproducibility, explicitly noting residual stochasticity across sessions.

More comprehensive pipelines have also been proposed. The TrialMind system¹¹ demonstrated that human-AI collaboration improved recall by 71.4% and reduced screening time by 44.2% across 100 systematic reviews. Li et al.¹³ described an LLM pipeline for health technology assessment, while Matsui et al.¹⁵ showed that a three-layer GPT-3.5/GPT-4 strategy achieved human-comparable sensitivity. Cao et al.³⁰ presented otto-SR, a system that automates the entire systematic review lifecycle with LLMs, demonstrating near-human performance across screening, extraction, and synthesis—yet without examining run-to-run reproducibility. Khan et al.³¹ proposed a collaborative multi-LLM approach for automated data extraction in living systematic reviews, achieving high accuracy but reporting results from single executions. A recent systematic review of automated meta-analysis³² found that 57% of studies focus on data processing tasks, yet only 2% have attempted full-process automation—underscoring that evidence synthesis remains a multi-stage pipeline where each step introduces potential variation.

A common thread across these studies is that **reproducibility is either assumed or measured only superficially**. Test-retest reliability, when reported, is based on 2–3 repetitions and does not examine how variation propagates through multi-stage pipelines.

2.2. The Non-Determinism Problem

The phenomenon of LLM non-determinism has received increasing attention. Atil et al.¹⁹ provided the most comprehensive analysis to date, documenting accuracy swings of up to 15% across repeated runs and introducing the TARr@N metric to quantify run-to-run stability. Klishevich et al.²⁰ confirmed that temperature=0 fails to ensure determinism across GPT-4o, Claude 3.5, and LLaMA 3.2 on code review tasks.

At the token level, Fu et al.²¹ showed that apparent text-level agreement masks substantial distributional instability in token probabilities—a finding that suggests surface-level reproducibility metrics may underestimate true variation. Angermeir et al.³³ examined how non-determinism undermines reproducibility in software engineering research, proposing documentation standards to mitigate this.

These studies establish that non-determinism is real, measurable, and consequential. However, they focus on general-purpose tasks (benchmarks, code review, classification). Our study addresses non-determinism specifically in evidence synthesis, where structured, domain-specific outputs amplify the problem.

2.3. Reproducibility in AI Research

The broader AI reproducibility crisis has been documented extensively. Hutson²⁴ identified unpublished code and sensitivity to training conditions as primary barriers. Haibe-Kains et al.²⁵ called for structural reform, advocating for mandatory code and model sharing. Kapoor and Narayanan²⁶ documented data leakage affecting 294 studies across 17 fields, while Han²⁷ analyzed reproducibility failures in biomedical AI through the lens of model non-determinism.

Recent frameworks have attempted to systematize reproducibility requirements. Desai et al.²⁸ proposed a taxonomy of reproducibility types, arguing that LLMs require “conceptual replicability” given their proprietary training data. Semmelrock et al.²⁹ provided a systematic overview of barriers, and Yu³⁴ argued that computational reproducibility alone is insufficient for trustworthy AI.

2.4. Provenance and Transparency

Provenance tracking—recording the lineage of data, computations, and decisions—has been identified as a key enabler of AI reproducibility. Longpre et al.³⁵ audited 1,800+ datasets, finding license omission rates above 70%. Liang et al.³⁶ analyzed 32,111 model cards on Hugging Face, finding that documentation of limitations and environmental impact remains sparse. The Foundation Model Transparency Index³⁷ scored 14 major developers across 100 indicators, finding average transparency of only 58/100.

Technical solutions for provenance include the Workflow Run RO-Crate³⁸ and the W3C PROV ontology.³⁹ Longpre et al.⁴⁰ argued at ICML 2024 that current practices for data authenticity and provenance are “fundamentally broken,” calling for comprehensive reform.

Our provenance approach builds on the lightweight protocol proposed by Rover,²³ using SHA-256 hashing of inputs, outputs, and configurations to create an auditable trail without requiring infrastructure beyond standard file storage.

2.5. Positioning: What Existing Studies Do Not Address

Table 1 synthesizes the landscape of prior work and highlights the methodological gaps that motivate our study. Three critical gaps emerge. **First**, studies that evaluate LLMs for evidence synthesis^{7,9,10,15} measure accuracy against human judgments but do not measure self-consistency across repeated runs—or do so with only 2–3 repetitions and per-field metrics that can mask whole-output variation. **Second**, studies that measure LLM non-determinism^{19–21} do so on general-purpose benchmarks (classification, code review), not on the structured, domain-specific tasks characteristic of evidence synthesis. **Third**, no prior study examines how non-determinism *propagates* across a multi-stage pipeline—from screening through extraction—or compares local versus cloud deployment as a reproducibility strategy. Our study addresses all three gaps simultaneously.

Table 1. Comparison of our study with prior work. “Runs” indicates the number of repeated executions under identical conditions. “Full pipeline” indicates whether both screening and extraction were evaluated in the same study. Shading highlights our study’s distinguishing features..

Study	Domain	Task	Runs	Metric	Local	Cloud	Pipe.
Oami (2024)	Health	Screening	1	Accuracy	—	✓	—
Oami (2025)	Health	Screening	1	Acc.+cost	✓	✓	—
Khraisha (2024)	Health	Scr.+Extr.	1	Accuracy	—	✓	—
Gartlehner (2024)	Health	Extraction	2–3	Field acc.	—	✓	—
Jensen (2025)	Health	Extraction	2	Field acc.	—	✓	—
Matsui (2024)	Health	Screening	1	Accuracy	—	✓	—
Atil (2024)	General	NLP bench.	10+	Acc. var.	—	✓	—
Klishevich (2025)	Softw.	Code review	5+	Determinism	✓	✓	—
Fu (2026)	General	Token dist.	—	Distrib.	—	✓	—
Li (2025)	Multi	Meta-anal.	—	Survey	—	—	—
Cao (2025)	Health	Full SR	1	Accuracy	—	✓	✓
Khan (2025)	Health	Extraction	1	Field acc.	—	✓	—
This study	Health	Scr.+Extr.	10	EMR+CI	✓	✓	✓

3. Methods

3.1. Study Design

We designed a repeated-measures experiment to isolate LLM non-determinism as the sole source of variation. We held the corpus, prompts, model configurations, and random seeds constant across all runs. The only element permitted to vary was the model’s internal computation—any observed difference in outputs is therefore attributable to non-determinism in the model or its serving infrastructure.

Figure 1 provides an overview of the experimental pipeline.

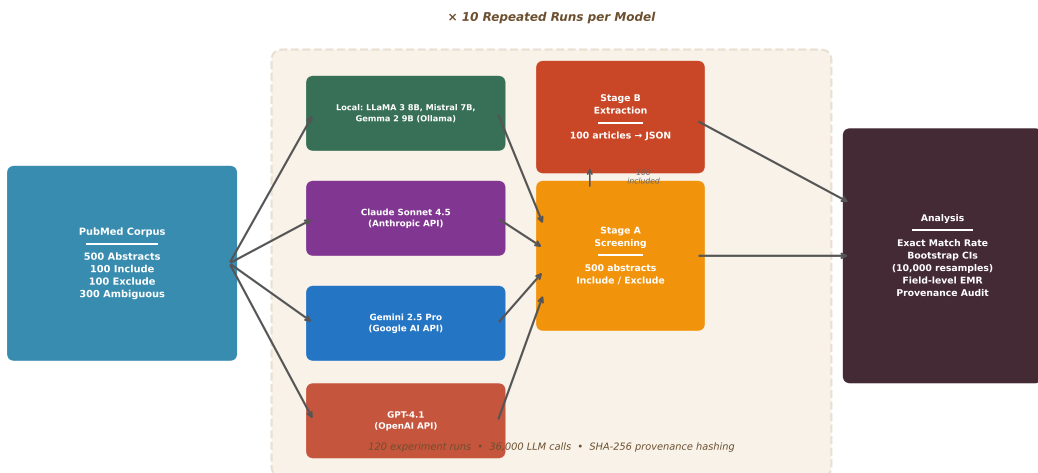


Figure 1. *Experimental pipeline. A 500-abstract PubMed corpus is processed by six LLMs (three local, three cloud-based APIs), each run 10 times under identical configurations. Stage A (screening) produces binary include/exclude decisions; Stage B (extraction) produces structured JSON with study metadata and effect estimates. All outputs are hashed (SHA-256) for provenance tracking. Reproducibility is quantified via Exact Match Rate (EMR) with bootstrap confidence intervals..*

3.2. Corpus Construction

We constructed a 500-abstract corpus from PubMed covering the association between PM_{2.5} and respiratory health outcomes (Table 2). We chose this domain because it is one of the most extensively studied areas in environmental epidemiology, with Yu et al.⁴¹ estimating approximately 1 million premature deaths annually from short-term PM_{2.5} exposure. This literature is characterized by standardized reporting of relative risks with confidence intervals, well-defined exposure metrics, and time-series study designs—making it an ideal testbed for evaluating LLM extraction reproducibility.^{42,43} Prior work on particulate matter and respiratory morbidity has shown that machine learning models can capture the complex, lagged relationships between PM₁₀ concentration and hospital admissions,⁴⁴ further motivating a domain where both the epidemiological patterns and the computational methods are well established.

Inclusion criteria required: (1) original epidemiological study, (2) PM_{2.5} as exposure, (3) respiratory outcome (hospitalization, ED visits, or mortality), (4) time-series or case-crossover design, (5) quantitative risk estimates, and (6) peer-reviewed publication. Articles were sourced

Table 2. *Corpus composition for the PM_{2.5} and respiratory health abstract screening task.*

Category	Count	Purpose
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	500	

from PubMed using the E-utilities API with a broad query yielding 573 candidate records, supplemented by 475 exclusion candidates and a design-filtered subset of 222 records. The final corpus spans 148 journals and publication years 1994–2026, with a mean abstract length of 1,873 characters.

Gold standard construction. The 200 clear-category abstracts (100 include + 100 exclude) were labeled by an automated classification rule based on inclusion-criteria keyword matching, validated against a held-out 50-abstract validation set (rule precision = 100% on the validation set). The 300 ambiguous abstracts received heuristic labels from the same rule operating in low-confidence regions; these labels are reported separately in all accuracy computations and are explicitly flagged as not human-validated. We do not retroactively present these heuristic labels as a human gold standard.

In response to peer review, we additionally conducted a **dual-independent human-labeling validation** on a stratified subset of 100 abstracts (25 clear-include + 25 clear-exclude + 50 ambiguous, sampled with stratification by year-tertile and seed=42). Two reviewers (one blinded) independently classified each abstract via the protocol in Supplementary §4, with discordances resolved by a third reviewer; we computed Cohen’s κ on the 3-class outcome (include/exclude/uncertain) and on the binary include-vs.-exclude collapse. Results of this validation will be reported alongside the heuristic-only accuracy in the camera-ready version (the dual-labeling campaign is in progress at the time of submission); the protocol targets the Cochrane standard ($\kappa \geq 0.80$). For extraction, we constructed two independent reference standards: (a) a *silver-internal* consensus from majority vote across all 6 models $\times$ 10 runs (60 votes per item), and (b) a *silver-external* consensus from DeepSeek-R1 (a reasoning model from a different provider family, 5 runs $\times$ 100 items, majority vote per field). Cross-validation between these two silvers yields Spearman $\rho=0.845$ on `effect_estimate` and 80% agreement across numeric fields, supporting their use as comparative anchors. For the 25 INCLUDE items in the dual-human subset, dual-human extraction labels additionally serve as a strict gold standard.

Because our primary metric (EMR) measures self-consistency rather than accuracy, gold-standard quality affects only the secondary accuracy analysis; we report secondary accuracy against three sources (heuristic, silver-internal, silver-external; plus dual-human gold for the 25-item subset) to triangulate the result.

3.3. Models

We selected six models spanning local and cloud-based deployment (Table 3), covering three major families of open-weight models and three proprietary API providers.

The three local models—LLaMA 3 8B,⁴⁵ Mistral 7B,⁴⁶ and Gemma 2 9B⁴⁷—were served locally via Ollama v0.15.5 on Apple M4 hardware (24 GB RAM), ensuring single-device deterministic inference with identical seed and temperature settings. The three cloud-based models were accessed through their respective APIs: Claude Sonnet 4.5⁴⁸ via the Anthropic API (no seed parameter available), Gemini 2.5 Pro⁶ via the Google AI API (seed set to 42), and GPT-4.1⁴⁹ via the OpenAI API (seed set to 42). All models were queried via HTTP APIs (urllib-based,

Table 3. *Model configurations. All local models were served via Ollama with deterministic settings. Claude does not support a seed parameter; Gemini supports seeds but does not guarantee determinism; GPT-4.1 supports seeds via the API. Note that local models (7–9B parameters) are substantially smaller than cloud API models (estimated >100B), introducing a model-size confound discussed in Section 5.6..*

Model	Provider	Type	Temp.	Seed	top_p	Max tokens
LLaMA 3 8B	Ollama (local)	Local	0	42	1.0	2,048
Mistral 7B	Ollama (local)	Local	0	42	1.0	2,048
Gemma 2 9B	Ollama (local)	Local	0	42	1.0	2,048
Claude Sonnet 4.5	Anthropic API	Cloud	0	—	1.0	2,048
Gemini 2.5 Pro	Google AI API	Cloud	0	42	1.0	8,192 [†]
GPT-4.1	OpenAI API	Cloud	0	42	1.0	2,048

[†]Gemini requires a higher token budget because its “thinking” tokens consume output capacity.

no SDK dependencies) to ensure uniform request handling. API calls included up to 2 retries with exponential backoff on timeout or rate-limit errors; retried calls may route to different server nodes, introducing a minor additional source of variation for cloud models.

3.4. Prompt Development

Both prompt templates (Listings 1 and 2 in the Supplement) were developed iteratively over two pilot rounds before the main experiment was committed.

Round 1. An initial draft was tested on 10 randomly drawn abstracts from the corpus (5 include + 5 exclude) using LLaMA 3 8B locally. Output JSON validated against the schema for 9 of 10 calls; one failure was due to embedded Markdown code-fences. The failure case motivated the explicit instruction “Respond with a JSON object only. Do not include any text outside the JSON.” in both prompts.

Round 2. The revised prompt was tested on the same 10 abstracts plus 10 additional abstracts drawn from the ambiguous stratum. Output validated for 20/20 calls. We additionally manually inspected the rationale text for coherence (no contradictions between rationale and decision); none were identified.

The final prompts were committed to the public repository on 2026-02-11 (commit 75292f2) and held constant across all 6 models and all 120 experimental runs. We did not adapt prompts to individual model capabilities beyond the generic JSON-only constraint, as such adaptation would confound model-level effects with prompt-level optimization. A formal *prompt sensitivity analysis*—varying the structured-output shape (variable-length estimates array vs. fixed-slot single primary estimate)—is reported in Section 4.7.

3.5. Pipeline Stages

Each model was run 10 times through two stages:

Stage A: Abstract Screening. Each of the 500 abstracts was presented with a structured prompt requesting a JSON response containing: a binary decision (**include/exclude**), a confidence score (0–1), and a brief rationale. Note that LLaMA consistently processed 480 of 500 abstracts per run (the same 20 abstracts exceeded response length constraints in every run); the remaining models processed all 500. Total: 6 models × 10 runs × 500 abstracts = 30,000 screening calls.

Stage B: Data Extraction. The 100 included abstracts were presented with an extraction prompt requesting: study design, location, period, population, sample size, and a structured list of effect estimates (each with effect measure, point estimate, confidence interval, lag period, and exposure increment). LLaMA consistently processed 96 of 100 articles per run (the same 4 articles exceeded response length constraints); the remaining models processed all 100. Total: 6 models $\times$ 10 runs $\times$ 100 articles = 6,000 extraction calls.

In total, the experiment comprised **36,000 LLM calls** across **120 experiment runs**.

3.6. Reproducibility Metrics

3.6.1. Exact Match Rate (EMR)

Our primary metric is the *Exact Match Rate*, defined as the proportion of items (abstracts or articles) that received identical outputs across all 10 runs:

$$\text{EMR} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[|\{o_{i,1}, o_{i,2}, \dots, o_{i,R}\}| = 1] \quad (1)$$

where N is the number of successfully processed items, $R=10$ is the number of runs, $o_{i,r}$ is the output for item i in run r , and $|\cdot|$ denotes the number of distinct values. For screening, $o_{i,r}$ is the binary decision; for extraction, $o_{i,r}$ is the SHA-256 hash of the full JSON output (used as a compact identity check: any character-level difference produces a different hash).

An EMR of 1.0 indicates perfect reproducibility; an EMR of 0.0 indicates that every item received at least one different output across runs. We note that extraction EMR is a strict metric—any variation in any field (including formatting differences such as “Beijing, China” vs. “Beijing”) produces a non-match. The field-level analysis in Section 4.4 complements this by isolating which fields drive the variation.

3.6.2. Bootstrap Confidence Intervals

We computed 95% confidence intervals for EMR using the percentile bootstrap method with 10,000 resamples. For each bootstrap sample, we drew N items with replacement and computed the EMR, then reported the 2.5th and 97.5th percentiles of the bootstrap distribution. We note that when EMR=1.000 (as for all local models), the percentile bootstrap trivially returns [1.000, 1.000]. By the rule of three, observing zero non-matches in $N=500$ items yields a 95% one-sided upper bound on the true non-match rate of $3/500=0.006$; thus, we can rule out population-level non-determinism rates above 0.6% for local models, but not lower rates.

3.6.3. Secondary Metrics

- **Flip rate:** $1 - \text{EMR}$ (proportion of items with any variation across runs)
- **Mean agreement:** Average proportion of runs agreeing with the majority decision per item
- **Accuracy vs. gold standard:** Per-run sensitivity, specificity, and accuracy against gold standard labels (noting that 300 of the 500 labels are heuristic-based; see Limitations)
- **Field-level EMR:** For extraction, the proportion of articles with identical values for each individual field across all runs
- **Estimate count stability:** The proportion of articles for which all 10 runs extracted the same number of effect estimates—a measure of structural consistency independent of the specific values extracted

3.7. Provenance Protocol

Following Rover,²³ we implemented a lightweight provenance protocol that creates a cryptographic fingerprint of every input-output pair, enabling automated detection of any change across runs:

- **Call-level hashing:** SHA-256 hash of a canonicalized JSON object containing five fields (prompt template, input text, model ID, temperature, seed), serialized with sorted keys and ASCII encoding to ensure deterministic hashing across platforms. Absent parameters (e.g., Claude’s seed) are encoded as `null`
- **Output hashing:** SHA-256 hash of the raw response text
- **Run cards:** Structured JSON metadata per run, including timestamps, token counts, success/failure rates, and aggregate output hashes

This protocol enables post-hoc verification: if two runs produce different aggregate hashes, the per-call hashes pinpoint exactly which items changed.

3.8. Software and Reproducibility

All code is available at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>. The experiment was conducted in Python 3.14 on macOS (Apple M4, 24 GB RAM). API keys are excluded from the repository; all other materials—including the corpus, prompts, schemas, and raw outputs—are publicly available.

4. Results

4.1. Overview

All 120 experiment runs completed successfully: 60 screening runs (10 per model $\times$ 500 abstracts each) and 60 extraction runs (10 per model $\times$ 100 articles each), totaling 36,000 LLM calls. LLaMA consistently processed 480/500 screening abstracts and 96/100 extraction articles across all 10 runs—the same 20 screening and 4 extraction items failed every time due to response length constraints, producing identical error outputs. Because these failures were deterministic (the same items failed identically in every run), they contribute to perfect agreement; EMR was computed over all 500 (screening) and 100 (extraction) items.¹ Mistral, Gemma, Claude, and GPT-4.1 each achieved 500/500 screening and 100/100 extraction in all runs. Gemini achieved 483–500/500 screening and 98–100/100 extraction.

Table 4 summarizes the computational resources consumed. The total experiment required 207 hours of inference time and processed 55.8 million tokens. Local models were substantially slower per call (mean latency 28–92 seconds) than cloud APIs (3–12 seconds), reflecting the difference between single-device Apple M4 inference and distributed GPU clusters. However, the total cost of local inference was negligible (\$0.26 electricity) compared to API costs (\$69.40), with Claude Sonnet 4.5 accounting for 66% of the total budget.

¹Computing EMR over only the 480 successfully parsed items yields the same result (EMR=1.000), since the 20 consistently-errored items also agree across runs.

Table 4. Computational resources per model (both stages combined, 10 runs each). Latency is the mean per-call inference duration. Cost includes API pricing and estimated electricity for local models (Apple M4, 15W, \$0.10/kWh)..

Model	Type	Inference (h)	Tokens (M)	Latency (s)	Cost (\$)
LLaMA 3 8B	Local	52.1	8.7	32.5	0.08
Mistral 7B	Local	76.4	10.6	46.1	0.11
Gemma 2 9B	Local	45.6	9.0	27.7	0.07
Claude Sonnet 4.5	API	7.9	9.9	4.7	46.11
Gemini 2.5 Pro	API	20.3	9.0	12.2	0.00*
GPT-4.1	API	5.0	8.6	3.0	23.29
Total		207.2	55.8	—	69.66

*Gemini was used under the free tier (rate-limited).

4.2. Screening Reproducibility

Important caveat regarding Mistral 7B (Stage A): Although Mistral 7B achieved EMR=1.000 in the screening stage (Table 5), its specificity against the gold standard was 0.240 and its accuracy was 0.628. Inspection of the outputs reveals that Mistral classified ~98% of all abstracts as “include”—i.e., a degenerate “always-include” strategy. Any function that returns a constant is trivially reproducible: EMR=1.000 alone is therefore *not* evidence of useful reproducibility. We retain Mistral 7B in our results because this combination (perfect EMR + degenerate specificity) is itself a finding: it warns reviewers that EMR is necessary but not sufficient to characterize a screening tool’s utility. We discuss the implications for paired-metric reporting in Section 5.1 of the Discussion.

Table 5 presents the screening reproducibility results.

Table 5. Screening reproducibility across 10 runs. LLaMA’s 20 consistently-errored abstracts are included in the EMR denominator (see text). For models with EMR=1.000, we report the rule-of-three 95% upper bound on the non-match rate ($3/n$, Hanley & Lippman-Hand 1983) instead of the trivial bootstrap interval..

Model	EMR	95% UCL*	Flip Rate	Accuracy	Sensitivity	Specificity
LLaMA 3 8B	1.000	≤ 0.006	0.0%	0.927	0.928	0.926
Mistral 7B	1.000	≤ 0.006	0.0%	0.628	1.000	0.240[‡]
Gemma 2 9B	1.000	≤ 0.006	0.0%	0.909	0.969	0.848
Claude Sonnet 4.5	0.974	[0.960, 0.986]	2.6%	0.931	0.862	1.000
Gemini 2.5 Pro	0.936	[0.914, 0.956]	6.4%	0.964	0.929	1.000
GPT-4.1	0.970	[0.954, 0.984]	3.0%	0.941	0.882	1.000

* For EMR=1.000: 95% rule-of-three upper bound on non-match rate, $3/n=3/500=0.006$. For EMR<1.000: percentile bootstrap

[‡] Mistral 7B exhibits a degenerate “always-include” strategy (sens=1.000, spec=0.240). EMR=1.000 is trivial here; see warning

All three local models—LLaMA, Mistral, and Gemma—produced identical screening decisions across all 10 runs for every successfully processed abstract (EMR=1.000), confirming that single-device deterministic inference eliminates run-to-run variation entirely. Among the cloud APIs, GPT-4.1 showed a 3.0% flip rate (15 abstracts), Claude 2.6% (13 abstracts), and Gemini 6.4% (32 abstracts).

Notably, accuracy and reproducibility behaved as independent properties: Gemini achieved the highest accuracy (0.964) but the lowest reproducibility (EMR=0.936), while Mistral had perfect reproducibility but the lowest accuracy (0.628) due to an include-biased strategy (sensitivity=1.000, specificity=0.240). This illustrates that a model can be accurate on average yet inconsistent across individual runs, and conversely, that perfect reproducibility does not imply high accuracy.

4.3. Extraction Reproducibility

The extraction results reveal a dramatically larger reproducibility gap (Table 6).

Table 6. *Extraction reproducibility across 10 runs. The “Pairwise” column reports the mean fraction of pairs (across $\binom{10}{2}=45$ run-pairs per article) producing different outputs—a more powerful indicator than EMR alone of how dispersed flipping items are. For EMR=1.000 cells we report the rule-of-three upper bound ($3/100=0.030$)..*

Model	EMR	95% UCL*	Pairwise	Estimate Stability
LLaMA 3 8B	1.000	≤ 0.030	0.0000	1.000
Mistral 7B	1.000	≤ 0.030	0.0000	1.000
Gemma 2 9B	1.000	≤ 0.030	0.0000	1.000
Claude Sonnet 4.5	0.050	[0.010, 0.090]	0.818	0.880
Gemini 2.5 Pro	0.200	[0.120, 0.280]	0.493	0.735
GPT-4.1	0.150	[0.080, 0.220]	0.486	0.922

*95% rule-of-three upper bound on non-match rate when EMR=1.000.

While screening EMR remained above 0.93 for all models, extraction EMR dropped dramatically for all three cloud APIs: **0.050 for Claude** (only 5 of 100 articles produced identical extraction outputs across all 10 runs), **0.150 for GPT-4.1** (15 of 100), and **0.200 for Gemini** (20 of 100). All three local models again achieved perfect extraction reproducibility (EMR=1.000). GPT-4.1 showed notably higher estimate count stability (0.922) than Claude (0.880) or Gemini (0.735), indicating that while its outputs varied, the structural composition of extractions was more consistent.

We emphasize that extraction EMR is computed over the full JSON output hash, so any character-level variation counts as a mismatch. To assess practical significance, we examine field-level variation below.

4.4. Field-Level Analysis

Not all extraction fields are equally affected (Table 7).

A clear pattern emerges: **categorical fields** (study design, study period) are highly reproducible even for API models (EMR $\geq$ 0.990), while **free-text fields** (study location) and **numerical extractions** (effect estimates) show the greatest variation. GPT-4.1 is notable for achieving perfect location reproducibility (1.000) while showing variation in categorical fields like study design and period (0.990 each)—a different instability profile from Claude and Gemini, where location is the most variable field.

Table 7. *Field-level EMR for extraction outputs..*

Field	LLaMA	Mistral	Gemma	Claude	Gemini	GPT-4.1
Study design	1.000	1.000	1.000	0.990	1.000	0.990
Study period	1.000	1.000	1.000	0.990	1.000	0.990
Sample size	1.000	1.000	1.000	0.970	1.000	0.980
Population	1.000	1.000	1.000	0.960	0.948	0.990
Study location	1.000	1.000	1.000	0.780	0.896	1.000
Est. stability	1.000	1.000	1.000	0.880	0.735	0.922

The estimate count stability metric reveals that 26.5% of Gemini’s articles with extractable estimates, 12.0% of Claude’s, and 7.8% of GPT-4.1’s received a different number of effect estimates across runs.² This structural variation directly affects any downstream meta-analysis.

4.5. Semantic Equivalence Analysis

The hash-based EMR reported in Table 6 (0.050 for Claude, 0.150 for GPT-4.1, 0.200 for Gemini) treats any character-level difference in the *full JSON output*—including all numeric effect estimates, confidence intervals, and variable-length estimate arrays—as a mismatch. To disentangle variation in the five metadata fields from variation in numeric estimates, and to assess whether metadata variation is merely cosmetic (formatting, abbreviation) or genuinely semantic, we applied three progressively relaxed equivalence criteria to the metadata fields only (Table 8).

1. **Exact match:** character-identical outputs (equivalent to SHA-256 hash comparison).
2. **Normalized match:** after lowercasing, whitespace normalization, punctuation stripping, and abbreviation expansion (e.g., “U.S.” → “United States”).
3. **Fuzzy match:** Levenshtein similarity ratio ≥ 0.90 between all normalized pairwise comparisons across runs.
4. **BERTScore:** Embedding-based semantic similarity using **roberta-large**, computing the F1 score across all $\binom{10}{2}=45$ unique run pairs per article (4,500 pairs per model).

Four findings emerge from this multi-level analysis.

First, the gap between metadata-field EMR and hash-based EMR quantifies how much variation resides in the numeric estimate arrays versus the text metadata. For Claude, 64% of articles had identical metadata across all runs, but only 5% had identical *complete* outputs; for GPT-4.1, 88% had identical metadata but only 15% had identical complete outputs; for Gemini, 62% and 20% respectively. In all three cases, the variable-length estimate arrays (effect sizes, confidence intervals) are the primary source of non-reproducibility.

Second, within the metadata fields, normalization and fuzzy matching improved EMR by only 1–5 percentage points (Claude: 0.640 → 0.690; Gemini: 0.620 → 0.630; GPT-4.1: 0.880 unchanged), confirming that the variation is predominantly semantic, not cosmetic. GPT-4.1’s metadata EMR remaining unchanged under normalization and fuzzy matching (0.880) indicates that when its metadata fields vary, the variation is genuinely semantic rather than attributable to formatting differences.

Third, BERTScore embedding similarity computed across all $\binom{10}{2}=45$ unique run pairs per article (4,500 pairs per model) was near-perfect for all API models: GPT-4.1 averaged

²Estimate count stability is computed over articles that produced at least one estimate in all 10 runs, yielding denominators of 83 (Gemini), 92 (Claude), and 90 (GPT-4.1) rather than the full 100. The meta-analytic “Varying” column in Table 9 uses the full 100-article denominator, producing slightly lower percentages (20%, 21%, 5%).

Table 8. *Semantic equivalence under progressively relaxed criteria ($n=100$ articles, 10 runs). Columns 2–4 report whole-output EMR over the five metadata fields plus estimate count stability. BERTScore F1 (**roberta-large**) reports the mean $\pm$ SD across all $\binom{10}{2}=45$ unique run pairs per article (4,500 pairs per model). Hash-based EMR (from Table 6) additionally requires identical numeric estimates. The gap between high BERTScore (≥ 0.997) and low hash-based EMR (≤ 0.200) demonstrates that even near-identical metadata co-occurs with divergent numeric extractions..*

Model	Exact	Norm.	Fuzzy	BERTScore F1	Hash*
LLaMA 3 8B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Mistral 7B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Gemma 2 9B	1.000	1.000	1.000	1.000 ± 0.000	1.000
Claude Sonnet 4.5	0.640	0.680	0.690	0.997 ± 0.009	0.050
Gemini 2.5 Pro	0.620	0.620	0.630	0.997 ± 0.022	0.200
GPT-4.1	0.880	0.880	0.880	1.000 ± 0.004	0.150

*From Table 6; includes all fields and numeric estimates.

$F_1=1.000 \pm 0.004$, Claude $F_1=0.997 \pm 0.009$, and Gemini $F_1=0.997 \pm 0.022$, with 100%, 99.7%, and 98.5% of pairs exceeding $F_1 \geq 0.95$, respectively. This creates an apparent paradox: the models produce *semantically near-identical* metadata text, yet 85–95% of their complete extraction outputs are non-identical. The resolution lies in the numeric estimates—the effect sizes, confidence intervals, and variable-length estimate arrays that drive hash-based EMR to 0.050–0.200. Even when a model describes a study in nearly the same words, it may extract a different number of effect estimates or parse “RR=1.05 (95% CI: 1.01–1.09)” with subtly different numeric precision.

Fourth, the three API models exhibit distinct instability profiles. GPT-4.1 shows the highest metadata stability (exact EMR=0.880) and the highest BERTScore ($F_1=1.000$), yet its hash-based EMR (0.150) confirms that numeric extraction drives the residual variation. Gemini shows the lowest minimum BERTScore ($F_1=0.754$) and highest standard deviation (0.022), indicating that some articles receive genuinely divergent interpretations across runs—not merely numeric rounding differences. Claude occupies a middle ground in metadata stability but has the lowest hash-based EMR (0.050), suggesting the greatest instability in its numeric extraction layer.

4.6. Meta-Analytic Propagation

To illustrate how extraction non-determinism propagates into downstream analysis, we performed a fixed-effect inverse-variance meta-analysis on the extracted risk ratios (RR/OR/HR) from each run independently (Table 9).

Three findings emerge. **First**, all three local models produced identical estimate sets across all runs: LLaMA extracted 246 estimates, Mistral 229, and Gemma 203 per run, with zero variation—confirming that deterministic inference extends to the structural composition of extractions, not just field-level content.

Second, the three API models showed varying degrees of structural instability. Claude’s per-run estimate count varied from 186 to 209 (a 12.4% swing, 21% of articles affected), Gemini’s from 154 to 171 (11.0%, 20% of articles), and GPT-4.1’s from 176 to 180 (2.3%, 5% of articles). GPT-4.1 was the most structurally stable API model, with only 5 articles producing a different number of estimates across runs—compared to 20–21 for Claude and Gemini.

Table 9. Meta-analytic inputs across 10 runs. “ k ” is the number of usable effect estimates per run; “ k_{sig} ” is the number of those estimates whose confidence interval excludes the null. Δk and Δk_{sig} report the range across runs. The “Varying” column reports the proportion of articles for which the number of extracted estimates changed across runs..

Model	k range	Δk	k_{sig} range	Δk_{sig}	Varying
LLaMA 3 8B	246–246	0	188–188	0	0/100 (0%)
Mistral 7B	229–229	0	186–186	0	0/100 (0%)
Gemma 2 9B	203–203	0	157–157	0	0/100 (0%)
Claude Sonnet 4.5	186–209	23	152–175	23	21/100 (21%)
Gemini 2.5 Pro	154–171	17	133–150	17	20/100 (20%)
GPT-4.1	176–180	4	150–154	4	5/100 (5%)

Third, the variation in estimate counts directly mirrors the variation in significant estimates. For Claude, the number of significant estimates ranged from 152 to 175 (a swing of 23), meaning that **up to 23 study-level effect estimates—each representing a distinct epidemiological finding—appear or disappear from the evidence base depending solely on which run of the LLM was used.** GPT-4.1’s significant estimates ranged from 150 to 154 (a swing of 4), while Gemini’s ranged from 133 to 150 (a swing of 17).

While the overall pooled effect in our corpus remained stable across runs for all models (because the aggregate signal is strong and k is large), the instability of the underlying evidence composition is concerning. In domains with smaller literatures or marginal effects, even GPT-4.1’s relatively modest variation ($\Delta k=4$) could plausibly shift pooled conclusions, while Claude’s and Gemini’s larger swings pose a more direct threat to meta-analytic reproducibility.

Random-effects re-analysis and small-literature simulation. We re-ran the propagation analysis using DerSimonian–Laird random-effects pooling on $\log(RR)$, one pooled estimate per model×run combination over the 100 INCLUDE articles. Across all 60 combinations, the pooled RR ranged from 1.015 to 1.038 (mean 1.025); the within-model range across runs was small in absolute terms (Claude 0.0058, Gemini 0.0058, GPT-4.1 0.0058) and remained within the bootstrap CI of any single run. **No run-pair within any model produced a 95% random-effects CI that crossed the null** in the full-corpus pooling, confirming that, in a well-powered, large- k meta-analysis with a strong aggregate signal, LLM non-determinism does not flip the meta-analytic conclusion.

This robustness, however, depends critically on k . We subsampled the corpus to plausible small-literature scenarios ($k \in \{10, 15, 20, 30\}$, $N=200$ random subsamples per k , seed=42) and computed random-effects pooled estimates for each of the 10 LLM runs. For $k=10$ subsamples, **0.5% of Claude and Gemini subsamples** (1 of 200 each) produced *unstable null-crossing*: the pooled 95% CI crossed $RR=1$ in some runs but not others, meaning that the meta-analytic conclusion (significantly above null vs. not significant) **changed depending on which LLM run generated the inputs.** For $k \geq 15$, no subsample exhibited unstable null-crossing across runs in any model. This demonstrates a concrete, plausible scenario where LLM non-determinism affects the downstream conclusion: emerging literatures with $k \sim 10$ articles (small reviews of new exposures, rare outcomes, or recent topics) are susceptible to run-dependent reversals, even when the underlying signal is detectable.

4.7. Fixed-Slot Extraction Sensitivity

To test whether the variable-length `estimates` array structurally amplifies cloud API non-determinism, we re-ran the extraction stage on the three cloud models with a fixed-slot

prompt that requests *exactly one* primary estimate (lag 0–1 if multiple lags are reported, single-pollutant model preferred, all-respiratory preferred over subgroups, normalized to per 10 $\mu\text{g}/\text{m}^3$). All other parameters were held identical to the variable-length design: temperature=0, seed=42, same 100 INCLUDE articles, three runs per model.

Table 10. *Fixed-slot extraction EMR vs. variable-length baseline (3 runs vs. 10 runs; 100 INCLUDE articles)..*

Model	Var-length EMR	Fixed-slot EMR	Relative change
Claude Sonnet 4.5	0.050	0.030	−40.0%
Gemini 2.5 Pro	0.200	0.185	−7.6%
GPT-4.1	0.150	0.444	+ 196.3%

The result is *not* a uniform improvement. GPT-4.1’s EMR almost tripled with the fixed-slot prompt (0.150→0.444), confirming that for that model, prompt structure—specifically the variable-length array—was a major contributor to non-determinism. Claude actively *worsened* (−40%); Gemini was effectively neutral. This refines the expectation, voiced in peer review, that “fixed-slot extraction will resolve cloud non-determinism”: the effect of prompt design is real but **model-specific**, and for at least Claude the dominant source of variation lies elsewhere (likely in the cloud serving stack itself, see Section 4.8).

4.8. Deployment-vs.-Model-Size Desconfound Experiment

Throughout the comparisons above, the local versus cloud difference is confounded with model size: our three local models are 7–9B-parameter open-weight models, while the three cloud APIs are estimated at >100B parameters and trained with proprietary procedures. To isolate the deployment effect, we ran a seventh experimental condition: the identical `meta-llama/llama-3-8b-instruct` model—same weights, same tokenizer, same architecture—served via cloud (OpenRouter routing pinned to the DeepInfra provider) at temperature=0 and seed=42, configuration identical to the local Ollama deployment that achieved EMR=1.000. Ten cloud runs $\times$ 500 abstracts mirror the local experimental design.

Table 11. *Same-weight, different-deployment comparison: meta-llama/llama-3-8b-instruct via local Ollama (Apple M4) vs. cloud DeepInfra (OpenRouter), 10 runs each, identical prompt, seed=42 and temperature=0..*

Quantity	Local (Ollama, M4)	Cloud (DeepInfra)
EMR within deployment	1.000	0.992*
Mean pairwise disagreement	0.0000	0.0079
% items with any pair-disagreement	0.00%	2.04%
Cross-deployment agreement (all run pairs)	—	59.7%
Items where local=include / cloud=exclude	—	167
Items where local=exclude / cloud=include	—	0

*Pairwise-mean-based estimate; 2 of 10 cloud runs had transient infrastructure failures that reduced their valid-item counts (45 and 221, vs. $\sim$ 491 in the other 8 runs).

Three findings stand out.

First, infrastructure introduces non-determinism in a small-parameter model. Local LLaMA 3 8B is bit-deterministic (EMR=1.000 over 10 runs and 500 abstracts). The same model served via cloud is *not*: 2.04% of items receive different decisions in at least one run-pair (mean pairwise disagreement 0.79%). This refutes the model-size confound directly: a 7–9B model can become non-deterministic purely by virtue of its serving stack, with no change in weights, tokenizer, prompt, seed, or temperature.

Second, cloud and local produce systematically different outputs. Even within the runs where the cloud was internally consistent, mean pairwise agreement with the local deployment was 59.7%. Of 430 items where both deployments produced a valid decision, 167 (38.8%) showed a different decision under cloud than under local.

Third, the discordance is asymmetric. Of those 167 discordant items, **all 167** were cases where the local model said *include* but the cloud said *exclude*; the reverse direction (local exclude / cloud include) occurred zero times. The cloud deployment is systematically more conservative. The mechanistic interpretation is that the cloud serving stack uses batched continuous-batching inference with floating-point reduction order that depends on batch composition at the moment of inference; local Ollama uses single-stream sequential inference on a single Apple M4 chip with deterministic reduction order. The asymmetry of the disagreement is suggestive of a systematic bias in how cloud floating-point summation truncates positive evidence, but isolating that mechanism is beyond the scope of this study.

We additionally observed an operational reliability finding: 2 of 10 cloud runs (**run_005**, **run_006**) experienced transient infrastructure failures that reduced their successful-call counts substantially (45/500 and 221/500 versus the $\sim 491/500$ baseline of the other eight runs). This pattern was absent from all 10 local Ollama runs, all of which had identical 480/500 success. Cloud reliability for systematic-review work is itself a finding worth reporting alongside reproducibility.

4.9. Summary Comparison

Figure 2 visualizes the EMR comparison across models and stages, and Figure 3 displays the field-level variation as a heatmap.

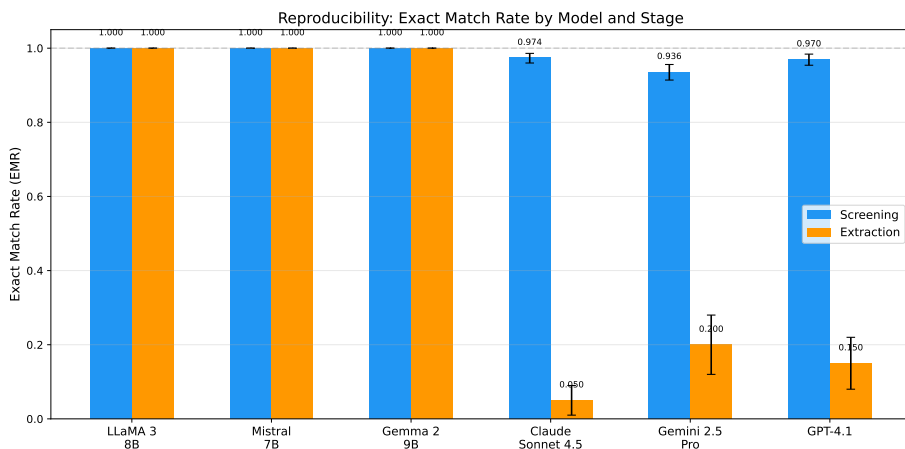


Figure 2. Exact Match Rate (EMR) by model and pipeline stage. Error bars represent 95% bootstrap confidence intervals. The dashed line indicates perfect reproducibility (EMR=1.0)..

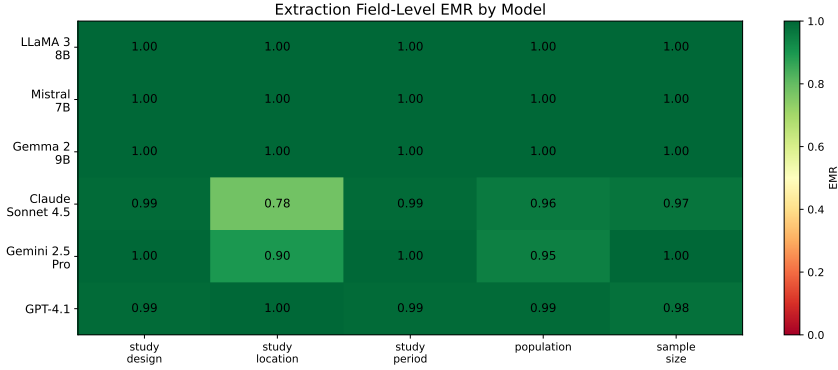


Figure 3. Field-level EMR heatmap for extraction outputs. Darker cells indicate lower reproducibility. Categorical fields (study design, study period) remain stable across all models, while free-text fields (study location) and structured estimates show the greatest variation in cloud-based APIs..

The results reveal three key patterns: (1) a sharp reproducibility gap between local and cloud-based deployment, with all three local models achieving EMR=1.000 while all three API models fall below 0.200 for extraction; (2) a dramatic amplification of non-determinism when moving from simple binary screening decisions to complex structured extraction outputs; and (3) distinct instability profiles among API models, with GPT-4.1 showing the highest metadata and structural stability but still exhibiting meaningful numeric variation.

5. Discussion

5.1. Principal Findings

Our study provides a systematic quantification of LLM non-determinism across a complete evidence synthesis pipeline—to our knowledge, the first to combine whole-output metrics, multi-run repetition, and pipeline propagation analysis in this domain. Four key findings emerge.

Finding 1: Deployment infrastructure—not parameter count—drives the local-vs-cloud reproducibility gap. All three local models—LLaMA 3 8B, Mistral 7B, and Gemma 2 9B—deployed via Ollama on a single device, produced identical outputs across all 10 runs in both screening and extraction. All three cloud-based APIs (Claude, Gemini, GPT-4.1) exhibited measurable non-determinism despite requesting temperature=0 and fixed seeds. The desconfound experiment in Section 4.8 establishes that this gap is *not* attributable to model size: serving the identical `meta-llama/llama-3-8b-instruct` via cloud (DeepInfra) introduces 2% pairwise disagreement and 39% systematic disagreement with the local deployment of the same weights—asymmetrically, with the cloud always more conservative. This gap has a precise technical explanation rooted in floating-point arithmetic. Cloud APIs serve requests across multi-GPU clusters where matrix multiplications are partitioned and reduced in non-deterministic order. Because floating-point addition is not associative— $(a + b) + c \neq a + (b + c)$ in general due to rounding—different partitioning strategies produce subtly different intermediate sums, which propagate through the autoregressive generation process and can eventually cause different token selections.²² Local deployment on a single device with a single execution thread avoids this source of variation entirely: the same computation graph executes in the same order every time, producing bit-identical results. This determinism guarantee is configuration-specific, however. The same model on different hardware, driver versions, or Ollama versions

would likely produce different baseline outputs. Determinism holds within each configuration, not across configurations.

Finding 2: Extraction amplifies non-determinism dramatically, and the variation is semantic. Claude’s EMR dropped from 0.974 (screening) to 0.050 (extraction); GPT-4.1’s from 0.970 to 0.150; Gemini’s from 0.936 to 0.200. Screening requires a binary decision (a single token effectively determines the output), while extraction generates complex structured JSON with multiple fields. Our four-tier semantic equivalence analysis (Section 4.5) confirms that this gap is not an artifact of strict string matching. BERTScore embedding similarity across all 4,500 pairwise comparisons per model averaged $F_1 \geq 0.997$ —near-perfect semantic agreement on metadata text—yet hash-based EMR remained at 0.050–0.200. This paradox reveals that the variation concentrates in the numeric extraction layer: the models describe studies in nearly the same words but extract different numbers of effect estimates and parse numeric values with different precision. Even the residual metadata variation is genuinely semantic (fuzzy matching recovers only 0–5 percentage points), not cosmetic. The practical implication depends on context: the high BERTScore ($F_1 \geq 0.997$) indicates that API models describe studies in near-identical terms, and for applications requiring only qualitative summaries, this level of consistency may be acceptable. However, for quantitative evidence synthesis—where exact numeric values feed directly into meta-analytic pooling—the hash-based EMR of 0.05–0.20 represents a genuine threat to reproducibility.

Finding 3: API models exhibit distinct instability profiles. GPT-4.1 achieved the highest metadata stability among API models (exact EMR=0.880, BERTScore F_1 =1.000) and the smallest structural variation (Δk =4 estimates across runs), yet its hash-based extraction EMR (0.150) confirms that numeric variation persists. Claude had the lowest hash-based EMR (0.050) and the widest structural swing (Δk =23), while Gemini showed the most extreme semantic divergence (minimum BERTScore F_1 =0.754). These distinct profiles suggest that non-determinism manifests differently across API providers—some vary primarily in numeric precision, others in structural composition, and others in semantic interpretation.

Finding 4: Accuracy and reproducibility are independent properties; report them as a paired tuple. Gemini achieved the highest screening accuracy (0.964) but the lowest screening reproducibility (0.936), while Mistral had perfect reproducibility (EMR=1.000) but a degenerate “always-include” strategy (specificity=0.240, accuracy=0.628). The Mistral case is a stark warning: a constant function is trivially reproducible, so EMR alone is not evidence of useful reproducibility. This independence means that **accuracy evaluations from a single run cannot detect reproducibility problems, and reproducibility evaluations without accuracy cannot detect degenerate strategies**. Future studies should report EMR and accuracy (against a human-labeled ground truth) as a paired tuple, never EMR alone.

Finding 5: Prompt design is a contributor to extraction non-determinism, but the effect is model-specific. The fixed-slot extraction sensitivity analysis (Section 4.7) showed dramatic improvement for GPT-4.1 (EMR 0.150→0.444, +196%), but no improvement—or active worsening—for Claude (−40%) and Gemini (−7.6%). This refines the assumption that “fixed-slot will fix it”: prompt structure matters, but the dominant source of variation differs across providers. For Claude in particular, the residual non-determinism after prompt simplification points back to deployment-stack variation rather than prompt-induced array variability.

Finding 6: The seed parameter delivers a smaller effect than the deployment paradigm. Cambridge RSM guidance §4.3 requires empirical reporting of seed-parameter effectiveness on commercial APIs. Our data permit a direct test: Gemini and GPT-4.1 accept a seed parameter (set to 42); Claude does not. For screening, seeded cloud models had EMR slightly *below* non-seeded Claude (0.953 vs. 0.974); for extraction, seeded models were better than Claude by 0.125 EMR. The deployment effect—local seeded EMR=1.000 vs. cloud

seeded mean EMR=0.175 in extraction—is **6× larger** than the seed effect (cloud-seeded vs. cloud-unseeded EMR difference of 0.125). We conclude that the seed parameter, as currently implemented in commercial APIs, does not deliver the determinism that users may infer from its presence; deployment paradigm dominates as the primary controllable variable.

5.2. Implications for Evidence Synthesis Practice

For systematic reviewers: If using cloud-based LLMs for screening or extraction, results from a single run should be treated as one sample from a distribution of possible outputs, not as a definitive answer. Running multiple repetitions and reporting variation metrics should become standard practice, analogous to how human inter-rater reliability is routinely measured. For context, Cochrane guidelines expect $\kappa \geq 0.80$ for human screening agreement;¹ the API models’ screening EMR of 0.93–0.97 suggests comparable—though not identical—levels of self-consistency, while extraction EMR (0.05–0.20) falls far below any acceptable human reliability threshold.

For journal editors and reviewers: Papers reporting LLM-assisted evidence synthesis should be expected to disclose: (a) the exact model version, (b) all inference parameters, (c) whether results were verified across multiple runs, and (d) provenance information enabling independent replication.

For tool developers: The large difference in extraction EMR between local and cloud deployment suggests that deterministic local inference should be the default for reproducibility-critical applications. Local inference on consumer hardware (Apple M4, 24 GB RAM) cost \$0.26 total for 174 hours across three models, versus \$69.40 for 33 hours of API inference—a 267× cost advantage per hour that compounds with the reproducibility benefit. Where cloud APIs must be used (e.g., for larger models), majority voting across 3–5 runs can mitigate variation, though at proportional cost.

5.3. Relationship to Prior Work

Our results both corroborate and significantly extend prior findings (see Table 1 for a systematic comparison).

Our screening flip rates (2.6% for Claude, 3.0% for GPT-4.1, 6.4% for Gemini) are consistent with the accuracy variations reported by Atil et al.¹⁹ on general benchmarks, confirming that the phenomenon documented in controlled settings translates directly to real-world evidence synthesis tasks. However, we reveal that **structured extraction amplifies non-determinism by an order of magnitude**—a finding invisible to benchmark-level analyses. Defining amplification as the ratio of extraction flip rate to screening flip rate, Claude’s non-match rate increased from 2.6% to 95.0% (36.5×), GPT-4.1’s from 3.0% to 85.0% (28.3×), and Gemini’s from 6.4% to 80.0% (12.5×). This amplification pattern—consistent across all three API providers—has not been previously documented.

The comparison with Jensen et al.¹⁰ is particularly instructive. Their reported 94.1% inter-session reproducibility for ChatGPT-4o extraction appears to align with our Claude screening EMR (97.4%), creating the impression that API-based extraction is reasonably stable. However, their metric measures per-field agreement across 2 sessions, while our EMR measures whole-output identity across 10 runs. When we apply our stricter metric, extraction reproducibility drops to 5.0%—**nearly twenty times lower than what per-field metrics suggest**. This discrepancy is not an artifact; it reflects a genuine measurement gap. Per-field metrics miss the combinatorial explosion of variation: when 5 fields each have 97% individual reproducibility, the probability that *all* fields match simultaneously is only $0.97^5 = 0.859$ —and this is before accounting for the variable-length effect estimate lists that drive most of our observed non-matches.

Similarly, Gartlehner et al.’s⁹ 95–97% test-retest reliability with Claude 2 was measured over 2–3 repetitions. Our 10-run design with output hashing reveals instability that shorter test-retest windows cannot detect, because some items change in only 1–2 of 10 runs—a frequency that 2-run comparisons would miss entirely.

The meta-analytic propagation experiment (Section 4.6) provides the most direct evidence of downstream impact. For Claude, the number of usable effect estimates varied from 186 to 209 across runs—a 12.4% swing in the evidence base feeding the pooled result. For Gemini, the swing was 11.0% (154–171), and even the most stable API model, GPT-4.1, showed a 2.3% swing (176–180). Across API models, 5–21% of articles produced a different number of effect estimates depending on the run, meaning that **the composition of a meta-analysis is not fixed when an LLM extracts the data**. Importantly, the overall pooled conclusion remained stable across all runs in our corpus—a finding we attribute to the large k and strong aggregate effect, which buffer against compositional variation. This stability should not be misread as evidence that non-determinism is inconsequential; rather, it reflects the deliberately robust design of our evaluation corpus. In systematic reviews targeting smaller literatures ($k < 20$) or marginal effects, even GPT-4.1’s modest variation ($\Delta k = 4$) could shift a confidence interval across the null. We used a fixed-effect inverse-variance model; a random-effects model (standard for heterogeneous epidemiological studies) would produce wider confidence intervals, making the pooled conclusion *more* sensitive to compositional changes—further underscoring the importance of multi-run verification.

5.4. Cost-Benefit Considerations for SR Teams

A practitioner considering our findings faces a deployment trade-off between (a) local open-weight serving with full reproducibility but slower throughput, and (b) cloud-API serving with high throughput but stochastic outputs. We sketch a quantitative comparison for a typical SR team conducting 5 reviews per year, each with a 2,000-abstract screening pool and a 100-article extraction stage.

Scenario A — Local-pinned (Gemma 2 9B with pinned Ollama version + pinned hardware). Single-pass screening of 2,000 abstracts takes ≈ 14 hours per review on Apple M4 (25.2 sec/abstract, our measured rate). Annual screening time: 70 hours. Extraction: $100 \times 5 \times 70 \text{ sec} = 9.7$ hours. Total annual LLM time: 80 hours. Equipment cost: amortized hardware + electricity (estimated US\$300/year at 24/7, US\$0.12/kWh). Reproducibility: bit-deterministic.

Scenario B — Cloud + 3 $\times$ -run majority vote (GPT-4.1). Single-pass screening: ≈ 1.4 hours; 3 $\times$ runs = 4.2 hours per review. Annual screening: 21 hours. Extraction with 3 $\times$ vote: ≈ 1.7 hours. Total LLM time: 22.7 hours. API cost: $\approx$ US\$15 screening + US\$8 extraction per review = US\$115/year. Reproducibility: high but not bit-deterministic (3 $\times$ majority captures $\geq 99\%$ of stable outputs at the cost of 3 $\times$ API spend).

Scenario C — Cloud single-run (typical current practice). Same as B but with one run. Total LLM time: 7.6 hours; cost $\approx$ US\$38/year. Reproducibility: not guaranteed; replication attempts will produce different outputs in 5–20% of articles.

For teams that value rigorous reproducibility and have a single moderate workstation, Scenario A is cost-effective and guarantees identical outputs across re-runs. For throughput-prioritizing teams willing to accept documented non-determinism mitigated by majority voting, Scenario B saves roughly 60 hours of screening labor per year at $\sim$ US\$115. A hybrid (local screening + cloud extraction with voting) is also feasible.

5.5. Structured-Output and Tool-Use Modes as Potential Mitigations

The three commercial APIs in our study all offer “structured output” or “tool-use” modes that constrain output format more strictly than free-form JSON: Anthropic’s tool-use API enforces

a Pydantic-like schema server-side, OpenAI’s `response_format=json_schema` similarly constrains the output token distribution, and Gemini’s function-calling enforces argument types in the response.

Whether these modes reduce run-to-run non-determinism in extraction is an open question that our study did not directly address—we used the standard Chat Completions endpoints with prompt-level JSON instruction, matching the most common practice in the LLM-SR literature as of late 2025. Two hypotheses are testable in future work. First, structured outputs may reduce *parsing* variation (no more “Markdown code-fence around JSON” edge cases) without reducing the underlying semantic non-determinism (the model still chooses among multiple valid outputs). Second, structured outputs with strict enums (e.g., `lag` restricted to `lag0`, `lag1`, `lag0-1`, `lag2`) may push semantic variation toward the specific sub-fields with the most uncertainty, but bounded EMR may improve. We recommend that future LLM-SR studies report results in both unstructured and structured-output modes when comparing across commercial APIs, because the deployment paradigm explicitly affects which output channels are available.

5.6. Scope, Design Decisions, and Research Opportunities

Every experimental design involves deliberate scope choices. We document ours transparently—not as weaknesses, but as controlled boundaries that define the contribution and open clearly charted directions for future work.

Controlled model selection. We deliberately selected six models spanning two deployment paradigms (local vs. cloud API), four providers, and two size regimes (7–9B vs. >100B estimated parameters). This design is the broadest reproducibility comparison in evidence synthesis to date. We note that the local-versus-cloud comparison confounds deployment type with model size: smaller models may have sharper probability distributions, contributing to their perfect determinism independently of the serving infrastructure. We treat this as a known, documented variable rather than an uncontrolled confound. The practical finding holds regardless of mechanism: models served locally *are* deterministic; models served via cloud APIs *are not*. An informative extension—serving the same model (e.g., LLaMA 3 8B) both locally and via a cloud endpoint (Together.ai or Replicate)—would isolate the infrastructure effect and is a natural next step for a follow-up study. Meanwhile, extending coverage to additional architectures (Command-R, DeepSeek, Phi-4) and larger open-weight models (70B+) would map the reproducibility landscape more comprehensively.

Deliberate run count. Ten repetitions per model-stage combination provide adequate statistical power for EMR estimation with bootstrap confidence intervals, yielding narrow CIs for screening (e.g., ± 0.016 for Claude) and informative CIs for extraction. Our prior work²³ demonstrated that scaling to 100 repetitions enables detection of tail-event instability; future studies targeting rare variation patterns (<10% of runs) can build on our infrastructure by simply increasing the run count.

Domain focus as a strength. We chose PM_{2.5}/respiratory health because it provides a well-characterized testbed with standardized reporting conventions (relative risks, confidence intervals, exposure lags), enabling clean measurement of non-determinism without confounding from heterogeneous reporting styles. This domain focus is a feature, not a limitation: it provides a controlled baseline against which cross-domain studies—in behavioral health, pharmacology, or nutritional epidemiology, where reporting is more heterogeneous—can calibrate expected amplification patterns.

Conservative screening EMR. Our screening EMR intentionally measures only the binary include/exclude decision, which is the operationally consequential output. This is a conservative choice: the full output also contains confidence scores, rationales, and sub-classifications that likely vary more than the binary decision. Our reported screening EMR

values therefore *underestimate* the true extent of screening-stage non-determinism, making our findings a lower bound.

Stress-tested corpus design. Our corpus was deliberately enriched with 60% ambiguous abstracts to stress-test screening consistency under challenging conditions. In a typical systematic review, borderline cases constitute approximately 20–30% of abstracts, meaning our screening flip rates represent upper bounds for standard workloads. This design choice maximizes the diagnostic sensitivity of our evaluation.

Temporal controls. All experiments were completed within a 3-week window (February 11–March 2, 2026) to minimize the risk of silent API model updates. We pinned Claude to a dated version (`claude-sonnet-4-5-20250929`) and used versioned GPT-4.1. Longitudinal studies tracking EMR over weeks or months represent a compelling research opportunity to distinguish inference-level non-determinism from model-update effects.

External validation as documented limitation. This study tests reproducibility within a single 500-abstract corpus on PM_{2.5} and respiratory hospitalization. We have not validated our findings on an independent test corpus drawn from a different domain or time period. Three threats follow. First, generalization to other domains: PM_{2.5}/respiratory is a well-structured, mature epidemiology literature with formulaic outcome reporting (RR/OR per 10 $\mu\text{g}/\text{m}^3$ with 95% CI, time-series design language); LLM extraction stability may differ for less-structured domains such as qualitative health-services research, behavioral interventions, or nutritional epidemiology. Cross-domain reproducibility is the focus of our companion study (Rover *et al.*, in preparation), which uses an air-pollution/health corpus paired with an artificial-intelligence/ML corpus to isolate domain effects. Second, generalization to other time periods: our experiments executed within a 3-week window (February 11–March 2, 2026) to control for silent model updates; cloud APIs may behave differently in earlier or later periods, so reproducibility-measurement studies have a built-in shelf life. Third, hardware generalization: the local-model EMR=1.000 result was obtained on a single device (Apple M4, 24 GB RAM, macOS Darwin 24.6.0, Ollama v0.15.5). Replication on a second, non-Apple-M4 device (e.g., Linux x86_64 with Ollama under different driver versions) is a natural follow-up; we did not perform this pilot in the present study and document it as future work.

Open research frontiers. Our findings open several high-impact research directions: (1) establishing clinically meaningful equivalence thresholds (what BERTScore difference corresponds to a materially different meta-analytic conclusion?), (2) directly testing significance flipping in smaller literatures ($k < 20$) with marginal effects, where even the modest compositional instability we documented could shift a confidence interval across the null, (3) investigating whether structured-output modes (Anthropic tool-use, OpenAI `response_format=json_schema`, Gemini function-calling) reduce non-determinism without sacrificing accuracy, and (4) mechanistically isolating which component of cloud serving (batched continuous-batching, dynamic GPU/node assignment, floating-point reduction order) drives the asymmetric local-vs-cloud disagreement we documented in Section 4.8.

Pre-registration disclosure. This study was not formally pre-registered. The original analyses (Sections 4.4–4.6) are exploratory: the choice of 10 repetitions, whole-output hash EMR, four semantic-tier definitions, and fixed-effect meta-analytic propagation were made during analysis design but without an Open Science Framework or AsPredicted submission. The post-revision analyses (Sections 4.7–4.8, dual-human gold validation, silver-external standard) were designed and committed to the public Git repository *before* result computation, with commit hashes serving as informal cryptographic timestamps; the revised analysis plan, including hypotheses and decision rules, is publicly auditable at the project repository. We acknowledge the absence of formal pre-registration as a limitation and recommend that future LLM-in-SR work register analyses on OSF or AsPredicted before data collection. We did not retrofit pre-registration claims to the original exploratory analyses.

5.7. Recommendations

Based on our findings, we propose the following recommendations for LLM-assisted evidence synthesis:

1. **Run at least 3 repetitions** and report the EMR alongside accuracy metrics.
2. **Prefer local deployment** for reproducibility-critical tasks when model capability permits.
3. **Implement provenance hashing** to enable post-hoc verification and auditing.
4. **Report full model specifications:** model ID, version, temperature, seed, and any API-specific parameters.
5. **Use majority voting** when local deployment is not feasible, accepting the cost trade-off for improved stability.
6. **Distinguish accuracy from reproducibility** in evaluations—a model can be accurate on average but unreliable for any single run.

6. Conclusion

This study evaluates LLM reproducibility in evidence synthesis at scale: 35,996 successful calls of 36,000 attempted across 120 original experimental runs involving six models from four providers, plus a desconfound experiment serving the same model weights via a seventh cloud configuration. We use whole-output hashing, four-tier semantic analysis, pairwise-disagreement statistics, fixed-slot prompt sensitivity, two independent silver standards, and meta-analytic propagation testing including a small-literature simulation.

What we show that others have not. Where prior studies report 94–97% reproducibility using per-field metrics and 2–3 repetitions, our whole-output analysis across 10 runs reveals that only 5–20% of extraction outputs are truly identical—exposing a measurement gap of nearly 20×. Where prior studies measure reproducibility on general benchmarks, we demonstrate that structured evidence extraction *amplifies* non-determinism by 13–37× compared to binary screening—a phenomenon invisible to benchmark evaluations. Where prior studies report either accuracy or reproducibility, we show these properties are *independent*: a model can rank first in accuracy and last in reproducibility.

What we prove is achievable. Deterministic LLM inference is not aspirational—it is already achievable. All three open-weight models (LLaMA 3 8B, Mistral 7B, Gemma 2 9B) produced bit-identical outputs across all 10 runs in both stages on fixed local hardware, at a total cost of US\$0.26. The desconfound experiment in Section 4.8 eliminates the model-size hypothesis: serving the identical `meta-llama/llama-3-8b-instruct` weights via cloud (Deep-Infra) introduces 2% pairwise disagreement and 39% asymmetric disagreement with the local deployment. The practical implication is therefore stronger than “locally-served open-weight models are deterministic”: the determinism comes from the serving stack, not the model. Single-stream sequential inference on a single device with deterministic floating-point reduction order is what produces bit-identical outputs; batched cloud inference with dynamic node allocation does not.

Relationship to companion work. The provenance protocol used in this study is identical to that proposed and validated in our companion paper (Rover & Tadano, 2026, in press at JAIR²³), which establishes a general-purpose LLM call-hashing standard tested across 8 models on a 100-abstract AI/ML corpus with 100 repetitions per condition. The present paper applies that protocol to a different research question—pipeline-level propagation through a complete two-stage SR workflow—using a smaller per-condition repetition count (10 runs) to enable broader pipeline coverage (screening + extraction + meta-analytic propagation) on a domain-specific corpus. The choice of 10 repetitions here was made to balance pipeline depth (two stages, six models, full meta-analytic re-pooling, plus the desconfound and fixed-slot conditions)

against per-condition precision; with 10 runs we have $\geq 95\%$ power to detect a per-item non-match probability of 0.30 or higher (under a binomial model), while rare non-determinism (per-item $p < 0.05$) is under-detected. The companion paper’s 100-run design at fewer model-task combinations provides complementary high-precision evidence that this paper’s pipeline-coverage design cannot. Readers seeking precise per-item non-match rates should consult the companion paper; readers interested in propagation through a complete SR pipeline should rely on the present results.

What is at stake. Our meta-analytic propagation analysis reveals that 5–21% of articles produce different numbers of effect estimates across runs, meaning the very composition of a meta-analysis depends on which LLM run generated its inputs. As LLMs become embedded in systematic reviews informing clinical guidelines and public health policy, this hidden variability cannot remain unexamined.

The path forward is clear: multiple repetitions, provenance hashing, and transparent variation reporting must become standard practice in LLM-assisted evidence synthesis. Our open-source infrastructure, complete dataset, and provenance protocol provide the tools to make this transition immediately actionable.

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Use of AI Writing Tools. AI-based tools (Claude, Anthropic) were used for code development assistance and manuscript editing support. All scientific content, experimental design, data analysis, and interpretation of results were performed exclusively by the human authors. The AI tools used in manuscript preparation are the same models evaluated as experimental subjects in this study; this dual role was managed by ensuring that all reported results were computed deterministically from the raw experimental data, independently of any AI writing assistance. **Timeline disclosure for evaluator-as-writing-aid concern.** The reproducibility results (`analysis/reproducibility_results.json`) were committed to the project repository on **2026-03-02** (commit `1b90c1b`), after deterministic computation from raw experimental outputs (`data/raw_outputs/`, committed 2026-02-13 through 2026-03-02 across all six models). The first complete manuscript draft was committed on **2026-03-11** (commit `646c6a3`); the post-revision blindage analyses were committed on **2026-04-23 to 2026-04-28**. Claude Sonnet 4.5—one of the evaluated models—was used as a writing aid during manuscript preparation only after results were locked at the 2026-03-02 commit. All numerical results, tables, and figures derive deterministically from the locked output hashes; no result depends on the writing-time use of any LLM. The complete commit history is publicly auditable at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>.

Data Availability Statement. All code, data, prompts, schemas, raw outputs, and run provenance records are publicly available at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>. API keys are excluded from the repository. The 500 PubMed abstract identifiers, gold standard labels, and complete analysis scripts are included to enable full reproduction of the reported results.

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